Paleomagnetic results from the Lower Paleozoic of Ireland

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Paleomagnetic data are presented for a variety of Ordovician rocks from two regions in Ireland: the South Mayo Trough and some Ordovician inliers of east-central Ireland. Some of these data update previous preliminary results, while others provide new information. In some instances the original remanence direction has been obscured by subsequent remagnetization.

Although magnetic inclinations of the Ordovician results are mostly shallower than contemporaneous British results, poor precision precludes any reliable estimate of closure across this sector of the Proto-Atlantic ocean. Declination differences between the South Mayo Trough and British data indicate a 30° clockwise rotation of this tectonic block during the Middle to Upper Devonian.

Nous présentons des données paléomagmatiques d'une série de roches Ordoviciennes de deux régions d'Irlande: la vallée de South Mayo et deux fenêtres Ordoviciennes du centre-est de l'Irlande. Quelques-unes de ces données confirment des résultats préliminaires antérieurs alors que d'autres fournissent de nouvelles informations. A quelques reprises, la direction originale de la rémanence a été masquée par une nouvelle magnétisation.

Même si les inclinaisons magnétiques de ces données Ordoviciennes sont plus faibles dans la majorité des cas que les données britanniques semblables, le peu de précision obtenue rend impossible une évaluation sérieuse de la fermeture de ce secteur de l'océan Proto-Atlantique. Les déclinaisons obtenues dans la vallée de South Mayo et celles des régions britanniques semblables indiquent qu'il y a eu une rotation de 30° dans le sens des aiguilles d'une montre de ce bloc tectonique durant le Dévonien moyen à supérieur.

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Introduction

Lower Paleozoic paleomagnetic data mostly from England and Scotland have been used to show the existence of a Proto-Atlantic ocean of limited extent (Briden *et al.* 1973). Presently available data from Ireland are sparse and difficult to interpret. Poles reported from the Mweelrea ignimbrites of Killary Harbour, although internally consistent, do not conform to the locally inferred Ordovician geomagnetic field as based on English data (Deutsch and Somayajulu 1970; Morris *et al.* 1973). Other reported results from Ireland are either based on limited stability criteria, small collections, or give directions suggestive of remagnetization, and hence are of limited use. In this paper, a preliminary study previously reported from the Salrock intrusives is updated in the light of further sampling and laboratory treatment. Also, from the South Mayo Trough, new data are reported from two felsites that outcrop along its eastern margin.

Excluding the data from Connemara, the only other Lower Paleozoic data from Ireland are the preliminary poles from the Ordovician inliers of Counties Louth, Meath, and Kildare (Morris and Robinson 1971). Extensive resampling of the two inliers at Kildare and Portrane, together with new collections from Balbriggan and Grangegeeth, provide a better estimate of the local Ordovician remanence direction.

All remanence measurements were made on a PAR-SM2D magnetometer. AF demagnetization was used to isolate the direction considered representative of the geomagnetic field at the time of rock formation. Both stable end-point and maximum Fisher precision methods were employed as selection criteria. As will be seen, in many cases the mean directions are poorly defined, suggesting that either the geomagnetic field at the time of remanence acquisition was weak (or unstable), or that AF cleaning fails to isolate the original mean direction. The sections on Balbriggan and Portrane and on the Glensaul and Derry Bay Felsites highlight this latter problem. At Balbriggan and Portrane, the recorded remanence has been partially reset. This was associated with chemical alteration; recovery of the original mean direction is therefore impossible. Both intrusive bodies reported in the next section appear to carry two stable (to AF)

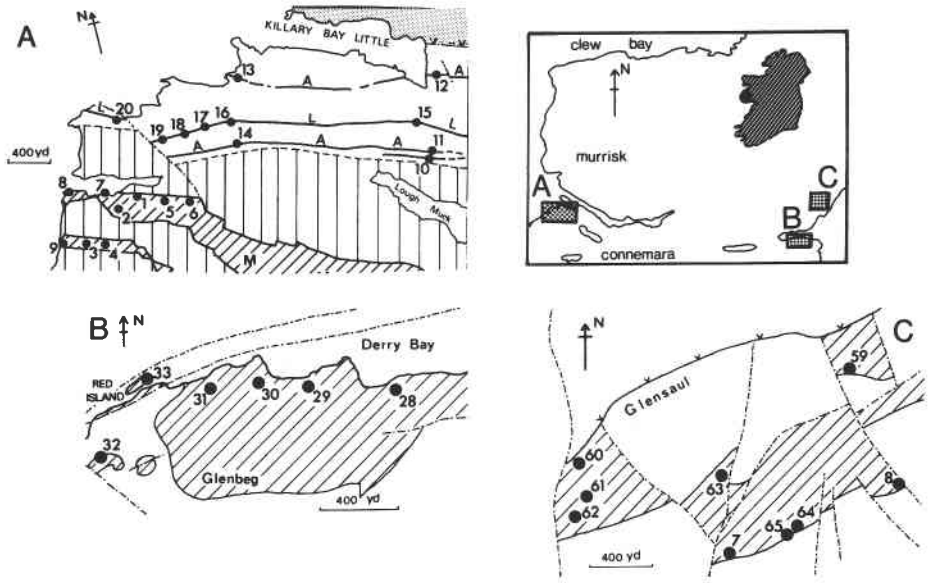


FIG. 1. Sampling locations in Western Ireland. The location of individual sampling regions within the South Mayo Trough are shown in the uppermost righthand diagram. (A) Sampling localities from intrusives in the Salrock (blank) and Upper Owenduff (vertical ruling) Groups; M—microgranodiorite, A—andesite, L—lamprophyre. Geology based on McKerrow and Campbell (1960). (B) Sampling sites in the Derry Bay Felsite (diagonal ruling). Structure and geology after Gardiner and Reynolds (1912). (C) Sampling sites in the Glensaul Felsite (diagonal ruling); geology after Gardiner and Reynolds (1910).

remanence directions. In each case, one direction can be interpreted in terms of the locally inferred geomagnetic field; the significance and relative age of the other direction is unknown. Moreover, accepting that reference to the horizontal plane represents the correct attitude at the time of extrusion (or intrusion) is merely a first approximation, and further compounds the difficulties of geological interpretation.

Glensaul and Derry Bay Felsites

The eastern margin of the South Mayo Trough is marked by the occurrence of a number of felsite masses. In every case, the local age relationships and mode of formation is debatable (Gardiner and Reynolds 1909, 1910, 1912). Most of the bodies are interpreted as being of intrusive origin; however, some exhibit contact relations with underlying beds more suggestive of extrusion. Cherts outcropping directly below the Glensaul body have a mid-Arenig fauna, similar to those interbedded with the Lough Nafooey spilites. Detailed mapping of individual flow units in the Glensaul region indicates an extrusive origin for this mass.

Stratigraphic relationships of the Derry Bay felsite are less clear; consequently, its age is questionable. Dewey *et al.* (1970), correlating this felsite with the Glensaul Group, interpreted formation as an ignimbrite sheet. On the other hand, W.E.A. Phillips (oral communication 1975), noting the complex contact relationships between the felsite and the overlying Silurian beds, considered an intrusive origin. More plausible, a Devonian age is further supported by the presence of small felsite dikes, which penetrate the Upper Llandovery sediments.

Nine sites in the Glensaul Felsite were collected by hand sampling; orientation was by magnetic compass (Fig. 1). After AF treatment, the results fall into two distinct groups (Table 1): Group A has a tilt-corrected mean based on five sites of $D = 197°$, $I = +48°$, while Group B has an *in situ* mean of $D = 353°$, $I = +75°$ ($n = 21$ samples). One of the Group A sites in the Glensaul collection is reversely magnetized with respect to the other four sites, indicative both of original remanence and an extrusive origin.

In the Derry Bay Felsite, four field-drilled sites were sampled from the margins of the main felsite mass, together with one from a fault sliver on Red Island and one from a small outlier on

TABLE 1. Glensaul felsite

Site	Optimum field	N (specimens)	R	k	α_{95}	D (*in situ*)	I (*in situ*)	D (dip corrected)	I (dip corrected)
			Site mean analysis—AF cleaned						
Group A									
59	300	4	3.273	4.1	52	200	+31	218	+63
64	300	7	5.517	4.0	34	187	−3	191	+27
65	300	5	4.321	5.9	34	207	−2	235	+43
7	100	6	5.786	23	14	176	+19	181	+62
8	300	5	4.566	9.2	27	350	−3	351	−32
Group B									
60	400	5	3.834	3.4	49	9	+79	332	+22
61	500	6	5.602	12	20	352	+77	330	+18
62	400	5	3.921	3.7	46	40	+74	340	+25
63	600	5	3.783	18	18	323	+65	340	+15

	N	R	k	α_{95}	D	I	Lat.	Long.
		Overall analysis						
Group A	(sites)							
in situ	5	4.730	15	21	188	+10	32 S	341 E
dip corrected	5	4.631	11	24	197	+48	7 S	335 E
Group B	(samples)							
in situ	21	17.891	6.4	14	353	+75	80 N	330 E
dip corrected	21	18.053	6.8	13	335	+20	43 N	205 E

NOTES: In all tables, the following conventions are followed: D and I are declination east and inclination positive downward; R is the resultant vector; k is the precision estimate; α_{95} is the radius of the circle of confidence $P = 0.05$. $d\psi/d\chi$—radii of the axes of the ellipse of confidence about the pole $P = 0.05$.

Kilbride peninsula (Fig. 1). After AF cleaning, all six sites have significantly grouped remanence (Table 2), with the directions again falling into two groups. As noted above, the age of the Derry Bay Felsite is uncertain. Accepting an Ordovician age of extrusion, the tilt-corrected result should agree with that from the Glensaul Felsite. Conversely, a Devonian intrusive event implies that the *in situ* direction should coincide with the defined local Devonian geomagnetic field. Since no crustal separation is envisaged in the Caledonide belt during the Devonian period, it is valid to compare this direction with Devonian data from England. Statistically, the *in situ* pole is identical to known British Upper Devonian results (Briden *et al.* 1973), while the tilt-corrected mean is quite different from the Glensaul felsite mean. Hence, an Upper Devonian age is preferred for the Derry Bay Felsite.

Both studies produced two directions of magnetization. In each case, there is no stratigraphic control over the location of sampling sites; hence, the temporal distribution of the differing directions is unknown. There are a number of possible ways to explain the Group B directions; these include remagnetization, polarity transition (especially for Glensaul), fault block rotation, and drilling-induced remanence. With the available data, it is not possible to differentiate between these options.

Upper Silurian Intrusives, Killary Harbour

In the region of Killary Bay Little, the local Wenlock sediments have been intruded by a number of sills. In the Upper Owenduff Formation there are two large microgranodiorite sills, while in the overlying Salrock Formation there are three sills of andesite and one of lamprophyre (Fig. 1A). Preliminary results from a small collection of these were previously reported by Morris *et al.* (1973).

Microgranodiorites

In this group there is a total of nine sites with no igneous contact tests. After AF cleaning, all sites had significantly grouped remanence (Table 3). The *in situ* mean of the nine sites is $D = 263°$,

TABLE 2. Derry Bay felsite

Site	Optimum field	N (specimens)	R	k	α_{95}	D (*in situ*)	I (*in situ*)	D (dip corrected)	I (dip corrected)
			Site mean analysis—AF cleaned						
Group A									
28	300	5	4.880	33	14	194	−9	194	+13
29	300	5	4.174	4.9	39	199	+18	203	+28
30	300	7	5.567	4.2	34	202	+47	217	+56
32	400	6	4.784	4.1	38	103	+51	162	+19
Group B									
31	200	6	5.073	5.4	32	119	+81	11	+80
33	50	6	5.357	7.8	26	13	+80	353	+69

	N (sites)	R	k	α_{95}	D	I	Lat. (paleomagnetic pole)	Long. (paleomagnetic pole)
		Overall analysis						
omitting sites 31, 33								
in situ	4	3.617	7.8	35	189	+35	28 S	341 E
dip corrected	4	3.617	7.8	35	192	+28	21 S	338 E

$I = +53$ ($k = 11$); after correcting for simple northerly tilt, the overall mean is $D = 318°$, $I = +44°$.

Andesites

All five sites with igneous contact tests are from two sections sampled in the enveloping Salrock Formation. The AF cleaned directions for the dike, the contact sediments, and the combined means are given in Table 3. Applying the test of Watson (1956) indicates that, at site 11, the contact and the intrusive samples have site means significantly different at the 95% level. Even though these contact samples may not have been in the baked zone, they were in the warmed zone. The difference in mean direction between the 'baked' contact and 'unbaked' sediment samples is much greater than that between the contact and the intrusive samples. Giving unit weight to each site, the *in situ* mean direction is $D = 252°$, $I = +20°$. Tilt corrected, this becomes $D = 290°$, $I = +46°$.

Lamprophyres

Only one lamprophyre sill cuts the two sections of the Salrock Formation; in each case, the site has an igneous contact test. Four other sites without contact tests were sampled in the sill where it crops out further to the west. After AF cleaning, all sites are significantly grouped (Table 3); the contact zones do not significantly differ from their adjacent intrusives. The *in situ* mean is $D = 240°$, $I = +38°$; tilt correction produces a mean of $D = 299°$, $I = +45°$.

Discussion

In the overall analysis, contact and igneous samples are treated as one site (Table 4). Statistically, the remanence directions from the three groups are not significantly different. This is not to say that the differing magmas were strictly contemporaneous; andesite dikes are seen to crosscut the microgranodiorite sills. The overall *in situ* mean direction is $D = 254°$, $I = +41°$; after tilt correction, this becomes $D = 306°$, $I = +45°$.

In order to establish which of these two directions is valid, it is necessary to elucidate the age of intrusion relative to folding of the enclosing Silurian sediments. McKerrow and Campbell (1960) show that the microgranodiorite follows a pre-existing NE–SW wrench fault that postdates folding of the Silurian. Further, minor support for post folding intrusion is provided by NRM directions in the Salrock Formation. The tilt corrected mean of this formation ($D = 45°$, $I = +1°$) most closely approaches opposing polarity to the *in situ* mean of the intrusives.

The Ordovician of County Meath and County Louth

To provide data from eastern Eire, samples were collected from three Lower Paleozoic inliers: the Chair of Kildare, Balbriggan and Portrane, and Grangegeeth (Fig. 2). From previous paleomagnetic collections reported from two of these inliers (Kildare and Portrane), Morris and Robinson (1971) concluded that the remanence

TABLE 3. Intrusives and baked contacts of Upper Silurian Age—Killary Harbour

Site	Optimum field	N	R	k	α_{95}	D (*in situ*)	I (*in situ*)	D (dip corrected)	I (dip corrected)
			Site mean analysis—AF cleaned						
Micrograndodiorite									
1	300	6	5.588	12	20	261	+57	316	+48
2	300	7	6.715	21	14	256	+30	283	+37
3	300	5	4.914	47	11	274	+17	291	+19
4	300	7	6.797	30	11	228	+48	317	+66
5	300	6	5.485	10	23	256	+61	327	+49
6	500	6	4.609	3.6	41	338	+58	353	+16
7	500	6	5.121	5.7	31	234	+44	294	+59
8	300	6	5.604	13	20	290	+68	342	+39
9	300	6	5.904	51	10	268	+60	328	+44
Andesites									
11	200	7	5.319	3.6	37	279	+3	290	+40
contact	400	3	1.851	not significant					
combined		10	7.544	3.6	29	280	+5	295	+35
12	200	7	6.736	23	13	306	+62	356	+9
contact	200	6	5.145	5.9	30	259	+39	348	+61
combined		13	11.293	7.0	17	280	+55	354	+31
13	600	7	5.322	3.6	37	249	+43	332	+58
contact	100	7	6.876	48	9	244	+11	272	+52
combined		14	11.573	5.6	19	246	+25	295	+43
14	500	7	6.573	14	17	227	+4	247	+43
contact	100	8	7.831	42	9	229	+19	266	+50
combined		15	14.284	20	9	228	+12	257	+47
15	200	6	4.898	4.5	35	257	−14	255	+16
contact	100	5	4.901	40	12	263	+4	280	+18
combined		11	9.600	7.1	18	260	−5	267	+17
Lamprophyres									
15	300	6	4.093	2.6	52	272	+35	311	+33
contact	300	4	3.748	12	28	249	+23	291	+46
combined		10	7.674	3.9	28	261	+30	302	+40
16	300	6	4.805	4.2	37	226	+58	315	+47
contact	100	6	5.829	29	13	226	+31	293	+43
combined		12	10.535	7.5	17	226	+44	302	+46
17	400	5	3.749	3.2	51	240	+22	260	+36
18	200	7	6.422	11	19	225	+39	301	+56
19	300	6	5.246	6.6	28	223	+73	350	+43
20	300	5	4.577	9.5	26	240	+21	280	+35

in the two groups was similar, and that both had been partially remagnetized and possibly rotated in the Permian.

The Chair of Kildare

Exposure of the volcanics in this inlier is restricted mainly to Grange Hill, as much of the surrounding area is drift covered. Only five sites were occupied, all hand sampled, and all oriented by magnetic compass. The volcanic units comprise andesites, porphyritic basalts, and tuffs (Gardiner and Reynolds 1896). Overlying this is a thick limestone unit which has yielded an Upper Ashgill fauna. Tectonic tilt is steep, uniform, and towards the southeast (Fig. 2).

Of the total NRM site means, only site 67 did not have significantly grouped remanence. Both the *in situ* and tilt-corrected overall means are nonsignificant. A pilot sample from each site was treated by progressive AF demagnetization after removal of an initial low coercivity VRM component; further treatment produced uniform intensity decay and direction change. An exception was the pilot sample from site 68, which exhibited little intensity decay (80% of NRM intensity was left after 1000 Oe treatment) and almost no direction change, suggestive that the remanence carrier may have been of the ilmenohematite series.

All sites were cleaned in 300 Oe or less, after

TABLE 4. Intrusives and Baked Contacts of Upper Silurian Age—Killary Harbour

	N (sites)	R	k	α_{95}	D	I	Lat. (paleomagnetic pole)	Long. (paleomagnetic pole)	$d\psi$	$d\chi$
Overall analysis—AF cleaned										
Microgranodiorite										
in situ	9	8.244	11	17	263	+53	23 N	286 E	16	23
dip corrected	9	8.275	11	16	318	+44	48 N	236 E	13	20
Andesites										
in situ	5	4.437	7.1	31	257	+19	1 S	276 E	17	32
dip corrected	5	4.345	6.1	34	293	+43	33 N	258 E	26	42
Lamprophyres										
in situ	8	7.485	14	16	240	+38	1 N	296 E	11	18
dip corrected	8	7.538	15	15	299	+45	38 N	254 E	12	19
Final analysis—AF cleaned										
Andesites, Lamprophyres, and Microgranodiorite Combined										
in situ	20	17.636	8.0	12	254	+41	9 N	286 E	9	15
dip corrected	20	18.035	9.7	11	306	+45	42 N	248 E	9	14

TABLE 5. The Kildare Inlier

Site	Optimum field	N (specimens)	R	k	α_{95}	D (*in situ*)	I (*in situ*)	D (dip corrected)	I (dip corrected)
AF cleaned—site mean statistics									
67	300	7	6.768	26	12	130	+40	173	+26
68	100	5	4.555	8.9	27	174	+31	184	−9
69	100	6	5.912	56	9	111	+29	155	+38
70	300	6	5.847	32	12	111	+47	178	+41
71	300	7	6.899	60	8	111	+61	195	+40

	N (sites)	R	k	α_{95}	D	I	Lat. (paleomagnetic pole)	Long. (paleomagnetic pole)	$d\psi$	$d\chi$
Overall analysis—AF cleaned										
in situ	5	4.642	11	24	129	+44	1 N	38 E	19	30
tilt corrected	5	4.642	11	24	177	+28	21 S	356 E	14	26

which they all exhibited significantly grouped remanence (Table 5). The *in situ* mean is $D = 129°$, $I = +44°$, whereas the tilt corrected mean is $D = 177°$, $I = +28°$ ($k = 11$).

Balbriggan and Portrane

At Balbriggan (Fig. 2), the shoreline outcrops of Ordovician volcanics are basic to intermediate andesites, which were apparently emplaced in a subaqueous environment (France 1967). Total thickness is variable, from 1500 m in the south to 40 m in the north. Tight folding and faulting of the volcanics provides a possible fold test of remanence. Three sites were also collected from Shenick's Island, where the Ordovician andesites are partly overturned (Bruck and Keenan 1970).

Volcanics also make up the major proportion of the rocks at Portrane, but here there is no evidence of submarine extrusion. After extensive alteration, the andesites are now mainly composed of calcite, with subsidiary chlorite and epidote. Other extrusive rocks exhibit large plagioclase phenocrysts (up to 1 cm long) in a fine-grained purple matrix, comparable to the nearby Lambay Porphyry (Gardiner and Reynolds 1898). Although the detailed structure of the Portrane inlier is complex, the lavas appear to be simply tilted to the east. AF cleaned site mean statistics are given in Table 8. Some sites

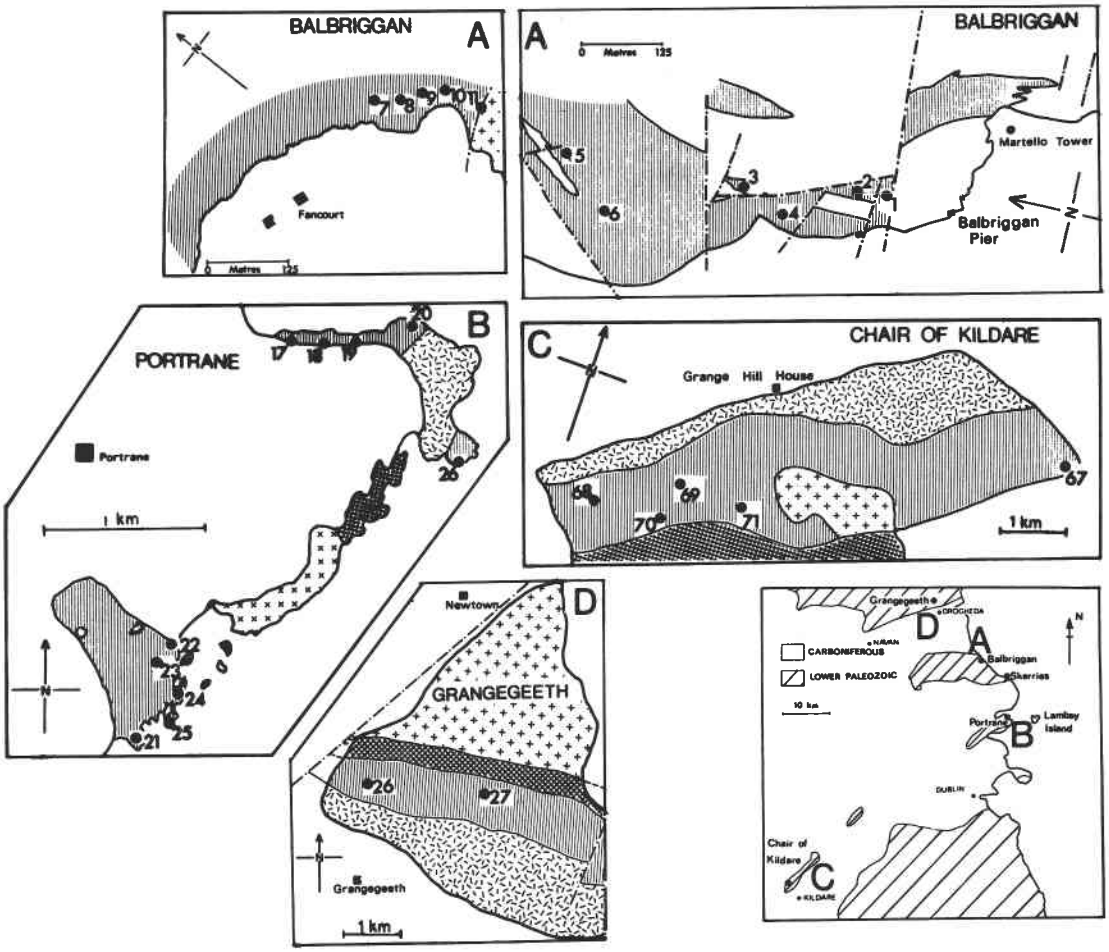


FIG. 2. Sampling locations in Eastern Ireland. The distribution of individual sampling regions is shown in the lowermost righthand diagram. (A) Sampling locations in andesites at Balbriggan; geology based on France (1967). (B) Sampling locations at Portrane; andesites shown by vertical ruling. Geology after Gardiner and Reynolds (1897). (C) Sampling locations in the chair of Kildare andesites (vertical ruling); geology after Gardiner and Reynolds (1896). (D) Sampling locations at Grangegeeth; geology after Harper (1952).

whose weak NRM intensity precluded any AF treatment are not considered further. The *in situ* AF cleaned mean of the Balbriggan results is not significant (Table 7). Tilt correction gives a significant mean of $D = 233°$, $I = -18°$ ($k = 4$). At Portrane both the *in situ* and tilt-corrected means are significant; the *in situ* mean is $D = 199°$, $I = -11°$ ($k = 4$), and the tilt-corrected mean is $D = 203°$, $I = +2°$ ($k = 3.8$).

Grangegeeth Volcanic Series

A large inlier of sediments and volcanics outcrops between the towns of Collon, County Louth, and Navan and Slane, County Meath (Fig. 2). The Volcanic Series consists of tuffs, lavas, and autobrecciated units (Harper 1952; Manistre 1952). Outcrop in the Llanvirn lavas is poor; hence, samples from only two sites were collected. Both were collected by drilling and oriented by sun compass. Prior to and after AF cleaning, the two sites have significantly grouped remanence (Table 8). Basing the mean on 12 AF-cleaned samples, the *in situ* mean is $D = 180°$, $I = +52°$; tilt correction modified this to $D = 193°$, $I = +26°$ ($k = 26$).

Summary of Data from Eastern Ireland

The Balbriggan and Portrane inliers yield

TABLE 6. AF cleaned, site mean statistics

Site	Optimum field	N (specimens)	R	k	α_{95}	D (*in situ*)	I (*in situ*)	D (dip corrected)	I (dip corrected)
Balbriggan									
1	500	5	4.912	45	12	297	−53	282	−13
2	300	5	4.858	29	14	237	−57	247	−14
3	400	6	5.386	8.2	25	197	+23	174	+28
4	300	4	3.747	12	28	208	−41	244	−33
7	200	6	5.573	12	20	206	−46	252	−35
8	300	6	4.819	4.2	37	201	−19	222	−22
11	200	6	4.538	3.4	43	203	+17	202	−13
Portrane									
17	300	6	5.321	7.4	27	224	+19	202	+31
18	100	6	5.624	13	19	228	−14	227	+17
19	300	6	4.586	3.6	42	193	−5	196	+7
20	200	6	5.949	120	6	146	+27	139	−4
21	200	6	5.881	42	11	145	+72	71	+54
22	400	6	5.772	22	15	192	0	190	+5
23	200	6	5.780	23	14	174	−28	196	−23
24	400	5	4.691	13	22	221	+21	195	+40
25	300	6	5.761	21	15	226	−44	246	−8

TABLE 7. Overall analysis—AF cleaned

	N (sites)	R	k	α_{95}	D	I	Lat. (Paleomagnetic pole)	Long. (Paleomagnetic pole)	$d\psi$	$d\chi$
Balbriggan and Shenick's Island										
in situ	7	5.445	3.9	35	214	−29	44 S	307 E	21	39
dip corrected	7	5.651	4.4	32	233	−18	29 S	291 E	17	34
Portrane										
in situ	9	6.843	3.7	31	199	−11	39 S	329 E	16	32
dip corrected	9	6.912	3.8	30	203	+2	32 S	326 E	15	30

TABLE 8. The Grangegeeth Volcanics

AF cleaned—site mean analysis

Site	Optimum field	N (specimens)	R	k	α_{95}	D (*in situ*)	I (*in situ*)	D (dip corrected)	I (dip corrected)
26	100	6	5.704	17	17	174	+46	186	+21
27	400	6	5.969	134	7	187	+57	199	+31

Overall analysis—AF cleaned

	N (specimens)	R	k	α_{95}	D	I	Lat. (Paleomagnetic pole)	Long. (Paleomagnetic pole)	$d\psi$	$d\chi$
in situ	12	11.582	26	9	180	+52	4 S	354 E	8	12
dip corrected	12	11.582	26	9	193	+26	22 S	340 E	5	9

comparable mean remanence directions, which are significantly different to other known Ordovician geomagnetic directions. Tilt correction as applied to the Balbriggan data produces an increase of precision, but the final mean still remains poorly defined. Plots of both the *in situ* and tilt corrected site mean directions exhibit a smeared distribution indicative of remagnetization (Fig. 3). This confirms the results of Morris and Robinson (1971), who, on the basis of their

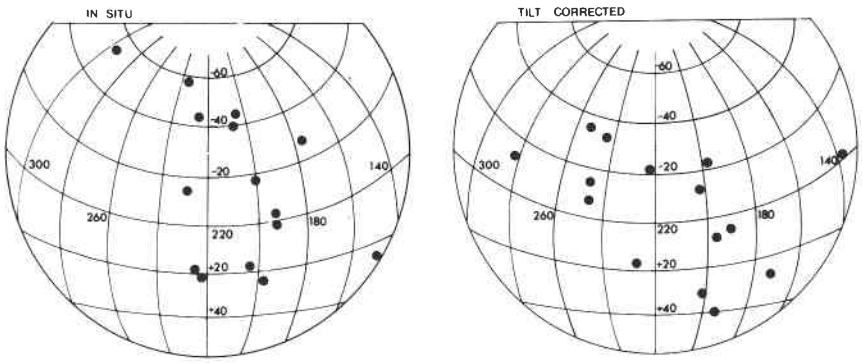


FIG. 3. Equatorial stereographic plots of site mean directions from Balbriggan and Portrane. Solid dots indicate outward directed remanence.

more limited data, argued for Permian remagnetization. Since neither mean direction has any geological significance, they are not considered further.

Results from the Kildare and Grangegeeth inliers have similar mean inclinations, but differ in magnetic declination by some 16°; this discrepancy is small, and may be related to the small size of these collections. Comparison with contemporaneous results from England indicates that these results are statistically compatible (Briden *et al.* 1973).

Conclusions

Previous paleomagnetic discussions on the extent of an Ordovician Proto-Atlantic ocean have been based on comparisons of data between Scotland and England (Briden *et al.* 1973), and North America and Africa (McElhinny and Opdyke 1973). The estimates given for total ocean closure in these two segments differ greatly: narrow in the Caledonide section, much wider in the Appalachian sector. In this paper, results are presented from two regions in Ireland that are considered to have lain on opposite margins of the proposed ocean (Dewey 1969). Geologically, the ocean is considered to have reached maximum extent during the early Ordovician. Hence, comparison of Ordovician paleomagnetic data from these two regions should provide an estimate of continental separation for this segment of the Caledonide belt.

Data pertaining to this discussion are: the Chair of Kildare and the Grangegeeth Volcanics from eastern Ireland, and the Glensaul 'felsite' and Mweelrea ignimbrite from the South Mayo Trough in the west of Ireland. Although the relevant poles are not strictly contemporaneous, their comparison remains valid. Data from other regions of the Caledonide–Appalachian belt indicate little or no polar wandering during the Lower to Middle Ordovician period.

None of the four relevant results presented here are well defined. In considering only magnetic inclination, the result from the Glensaul 'felsite' appears most similar to that in the Aberdeenshire gabbros. The other results apparently have systematically shallower inclinations; however, they are not significantly different to their relevant British counterparts. Magnetic inclination, therefore, does not provide any reliable estimate of ocean closure.

The pole reported from the Mweelrea ignimbrites remains anomalous, yet its repeatability verifies its validity (Deutsch and Somayajulu 1970; Morris *et al.* 1973). It was on the basis of this discrepancy that Deutsch (1969) argued for the existence of a wide Proto-Atlantic ocean for this portion of the Caledonide belt. This apparent difference can now be seen to result largely from a magnetic declination discrepancy. Ordovician results from the British Isles, both north and south of the proposed suture, exhibit little variation of magnetic declination, having a mean of approximately 170°. The magnetic declinations of the Glensaul and Mweelrea results differ from this by approximately 30°, *but* this difference is not in the same sense. The result from the Salrock intrusives (of late Silurian or early Devonian age), whilst yielding a magnetic inclination coincident with that from Lower Devonian studies in Scotland (Briden *et al.* 1973), differs significantly in magnetic declination. This difference is in the same sense and amount from the defined

Lower Devonian mean as the Glensaul result differs from the Ordovician mean. Explanation of the Mweelrea pole in terms of a wide Proto-Atlantic ocean is therefore untenable. Firstly, the Mweelrea and Glensaul studies belong to the same crustal segment, yet do not exhibit a unidirectional discrepancy. Second, the discrepancy continued until at least early Devonian times, when it was considered that ocean closure was complete. A new explanation must be sought in the geology of the South Mayo Trough.

The regional strike of Caledonide structures within the South Mayo Trough swings from the more normal NE–SW trend in the east to ESE–WNW in the west (Fig. 4). Rotating this anomalous tectonic trend back into alignment nullifies the 30° clockwise apparent declination discrepancy as seen in the results from the Glensaul Felsite and Salrock intrusives. It would appear, then, that the declination anomaly in the Mweelrea ignimbrites is of the order of 60°. Only the Mweelrea ignimbrites lie above the Salrock thrust, and it is to rotational movement during thrust development that this anomalous result is attributed.

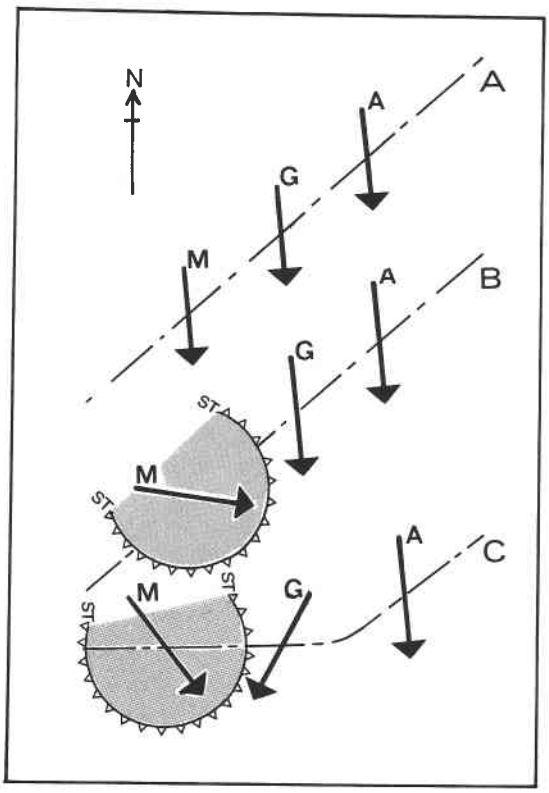


FIG. 4. Diagrammatical representation of the development of the South Mayo Trough. (A) Initial stage, L. Ordovician, all remanence declinations equal. (B) Development of Salrock thrust, rotation of Mweelrea ignimbrites, L. Devonian (?). (C) Rotation of South Mayo Trough, Middle to Upper Devonian. M—mean remanence declination in Mweelrea ignimbrites; G—mean remanence declination in Glensaul Felsite, A—average Ordovician declination for Britain. ST—Salrock Thrust. Dot dash line—axial trace of Caledonian structures.

Geologically, the age of this thrust is confined to the period post-Wenlock to pre-Carboniferous. Detailed studies of Devonian rocks in the nearby Ox Mountains date a similar thrust as post mid-Devonian (Currall 1963). It is probable that the thrusting predates the rotation of the South Mayo Trough, since structures on either side of the thrust are parallel. Paleomagnetically, this rotation is restricted to the period post-Lower Devonian to pre-Upper Devonian; the *in situ* pole from the Derry Bay Felsite is statistically identical to other known Upper Devonian results from the Caledonide belt.

In conclusion: these results do not provide any estimate of ocean closure for this segment of the Caledonide orogen because of poorly defined data points. Complex magnetizations were found in at least two studies, but with the techniques available, it was not possible to establish the relative stability or age of the two components. In each case, one direction could be correlated with the locally defined geomagnetic field for the specific time period represented by the rocks; the significance of the second direction is unknown. Explanation of the anomalous Mweelrea ignimbrite pole in terms of a large Proto-Atlantic ocean is now invalid. Rather, it is due to rotation of small tectonic blocks and thrust tectonics that accompanied the final welding of the Caledonian suture during the Middle to Upper Devonian.

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